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Operations Research Methods

Final Exam

Problem #1

Four products are manufactured on each of five machines. A pound of each product requires a specified number of hours on each machine, as presented in the following table. Total hours available on machine A, B, C, D, and E are 10, 16, 20, 18, and 12, respectively. The profit contributions per pound of products 1, 2, 3, and 4 are $4, $2, $3, and $6, respectively.

Machine	Machine-Hour Requirement			
	Product 1	Product 2	Product 3	Product 4
A	3	2	4	5
B	1	4	3	3
C	5	3	6	3
D	2	1	1	1
E	2	1	2	4

Formulate the linear programming model to determine how the manufacturer can maximize his profit. (15 pts)

Defining the variables

Let

x_1 = Amount of product produced (pounds)

x_2 = Amount of product produced (pounds)

x_3 = Amount of product produced (pounds)

x_4 = Amount of product produced (pounds)

Constraints

$3x_1 + 2x_2 + 4x_3 + 5x_4 \leq 10$

$x_1 + 4x_2 + 3x_3 + 3x_4 \leq 16$

$5x_1 + 3x_2 + 6x_3 + 3x_4 \leq 20$

$2x_1 + x_2 + x_3 + x_4 \leq 18$

$2x_1 + x_2 + 2x_3 + 4x_4 \leq 12$

$x_1, x_2, x_3, x_4 \geq 0$

Objective Function

$Z = 4x_1 + 2x_2 + 3x_3 + 6x_4$

Mathematical Model

Maximize $Z = 4x_1 + 2x_2 + 3x_3 + 6x_4$

Subject to:

$3x_1 + 2x_2 + 4x_3 + 5x_4 \leq 10$

$x_1 + 4x_2 + 3x_3 + 3x_4 \leq 16$

$5x_1 + 3x_2 + 6x_3 + 3x_4 \leq 20$

$2x_1 + x_2 + x_3 + x_4 \leq 18$

$2x_1 + x_2 + 2x_3 + 4x_4 \leq 12$

$x_1, x_2, x_3, x_4 \geq 0$

Problem #2

A corporation has decided to produce four new products. Five branch plants now have excess production capacity. The unit manufacturing cost of the first product would be $28, $24, $30, and $27 in plants A, B, C, and E, respectively, whereas, plant D does not have the capability for producing this product. The unit manufacturing cost of the second product would be $29, $33, $28, $32, and $31 in plants A, B, C, D, and E, respectively. The unit manufacturing cost of the third product would be $39, $32, $28, and $31 in plants A, B, D, and E, respectively, whereas, plant C does not have the capability for producing this product. The unit manufacturing cost of the fourth product would be $30, $26, $25, $32, and $31 in plants A, B, C, D, and E, respectively.

Sales forecasts indicate that 500, 300, 200, and 400 units of products 1, 2, 3, and 4, respectively, should be produced per day. Plant A, B, C, D, and E have the capacity to produce 300, 400, 200, 500, and 300 units daily, respectively, regardless of the product or combination of products involved. Management wishes to know how they should allocate the new products to plants in order to minimize total manufacturing cost.

a) Formulate this problem as a transportation problem by constructing the appropriate cost and requirements table.

	Product 1	Product 2	Product 4	Product 4		Supply
Plant A	28	29	39	30		300
Plant B	24	33	32	26		400
Plant C	30	28	--	25		200
Plant D	--	32	28	32		500
Plant E	27	31	31	31		300
Demand	500	300	200	400		

Note that the sum of supply = 1700. Also, the sum of demand = 1400. So, the difference between supply and demand is 1700 – 1400 = 300. Thus, a pseudo-product with a demand of 300 will be utilized for the remaining parts of the problem.

b) Use the northwest-corner method to obtain an initial basic feasible solution for the problem.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A						300
Plant B						400
Plant C						200
Plant D						500
Plant E						300
Demand	500	300	200	400	300	

Minimum{500, 300} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B						400
Plant C						200
Plant D						500
Plant E						300
Demand	200	300	200	400	300	

Minimum{200, 400} = 200

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200					200
Plant C						200
Plant D						500
Plant E						300
Demand	0	300	200	400	300	

Minimum{200, 300} = 200

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C						200
Plant D						500
Plant E						300

Demand	0	100	200	400	300	

Minimum{200, 100} = 100

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C		100				100
Plant D						500
Plant E						300
Demand	0	0	200	400	300	

Minimum{100, 200} = 100

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C		100	100			0
Plant D						500
Plant E						300
Demand	0	0	100	400	300	

Minimum{100, 500} = 100

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C		100	100			0
Plant D			100			400
Plant E						300
Demand	0	0	0	400	300	

Minimum{400, 400} = 400

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C		100	100			0
Plant D			100	400		0
Plant E						300
Demand	0	0	0	0	300	

Minimum{300, 300} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	300					0
Plant B	200	200				0
Plant C		100	100			0
Plant D			100	400		0
Plant E					300	0
Demand	0	0	0	0	0	

c) Use the minimum-cost method to obtain an initial basic feasible solution for the problem.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	28	29	39	30	0	300
Plant B	24	33	32	26	0	400
Plant C	30	28	--	25	0	200
Plant D	--	32	28	32	0	500
Plant E	27	31	31	31	0	300
Demand	500	300	200	400	300	

Minimum{28, 29, 39, 30, 24, 33, 32, 26, 30, 28, 25, 32, 28, 32, 27, 31, 31, 31, 0, 0 ,0} = 0

Minimum{300, 300} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	28	29	39	30	300	0
Plant B	24	33	32	26		400
Plant C	30	28	--	25		200
Plant D	--	32	28	32		500
Plant E	27	31	31	31		300
Demand	500	300	200	400	0	

Minimum{28, 29, 39, 30, 24, 33, 32, 26, 30, 28, 25, 32, 28, 32, 27, 31, 31, 31} = 24.

Minimum{400, 500} = 400

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	28	29	39	30	300	0
Plant B	400					0
Plant C	30	28	--	25		200
Plant D	--	32	28	32		500

Plant E	27	31	31	31		300
Demand	100	300	200	400	0	

Minimum{28, 29, 39, 30, 30, 28, 25, 32, 28, 32, 27, 31, 31, 31} = 25.

Minimum{200, 400} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	28	29	39	30	300	0
Plant B	400					0
Plant C				200		0
Plant D	--	32	28	32		500
Plant E	27	31	31	31		300
Demand	100	300	200	200	0	

Minimum{28, 29, 39, 30, 32, 28, 32, 27, 31, 31, 31} = 27.

Minimum{100, 0} = 0

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D	--	32	28	32		500
Plant E	27	31	31	31		300
Demand	100	300	200	200	0	

Minimum{32, 28, 32, 27, 31, 31, 31} = 27.

Minimum{100, 300} = 100.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D		32	28	32		500
Plant E	100	31	31	31		200
Demand	0	300	200	200	0	

Minimum{32, 28, 32, 31, 31, 31} = 28.

Minimum{200, 500} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D		32	200	32		300
Plant E	100	31		31		200
Demand	0	300	0	200	0	

Minimum{32, 32, 31, 31} = 31.

Minimum{200, 200} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D		32	200			300
Plant E	100	31		200		0
Demand	0	300	0	0	0	

Minimum{31, 32} = 31.

Minimum{300, 0} = 0.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D		32	200			300
Plant E	100	0		200		0
Demand	0	300	0	0	0	

Minimum{32} = 32.

Minimum{300, 300} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	0				300	0
Plant B	400					0
Plant C				200		0
Plant D		300	200			0
Plant E	100	0		200		0
Demand	0	300	0	0	0	

d) Use the Vogel's method to obtain an initial basic feasible solution for the problem.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply
Plant A	28	29	39	30	0	300
Plant B	24	33	32	26	0	400
Plant C	30	28	--	25	0	200
Plant D	--	32	28	32	0	500
Plant E	27	31	31	31	0	300
Demand	500	300	200	400	300	

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28	29	39	30	0	300	28
Plant B	24	33	32	26	0	400	24
Plant C	30	28	--	25	0	200	25
Plant D	--	32	28	32	0	500	28
Plant E	27	31	31	31	0	300	27
Demand	500	300	200	400	300		
Penalty	3	1	3	1	0		

Max{28, 24, 25, 28, 27, 3, 1, 3, 1, 0} = 28

Min{3, 1, 3, 1, 0} = 0

Minimum{300, 500} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28	29	39	30		300	1
Plant B	24	33	32	26		400	2
Plant C	30	28	--	25		200	3
Plant D	--	32	28	32	300	200	4
Plant E	27	31	31	31		300	4
Demand	500	300	200	400	0		
Penalty	3	1	3	1			

Max{1, 2, 3, 4, 4, 3, 1, 3, 1} = 4.

Min{3, 1, 3, 1} = 1.

Minimum{300, 300} = 300

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28		39	30		300	2
Plant B	24		32	26		400	2
Plant C	30		--	25		200	5
Plant D	--		28	32	300	200	4
Plant E	27	300	31	31		0	4
Demand	500	0	200	400	0		
Penalty	3		3	1			

Max{2, 2, 5, 4, 4, 3, 3, 1} = 5.

Min{3, 3, 1} = 1.

Minimum{200, 400} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28		39	30		300	2
Plant B	24		32	26		400	2
Plant C				200		0	
Plant D	--		28	32	300	200	4
Plant E	27	300	31	31		0	4
Demand	500	0	200	200	0		
Penalty	1		3	4			

Max{2, 2, 4, 4, 1, 3, 4} = 4.

Min{1, 3, 4} = 1.

Minimum{0, 500} = 0.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28		39	30		300	2
Plant B	24		32	26		400	2
Plant C				200		0	
Plant D	--		28	32	300	200	4
Plant E	0	300				0	
Demand	500	0	200	200	0		
Penalty	4		4	4			

Max{2, 2, 4, 4, 4, 4} = 4.

Min{4, 4, 4} = 4.

Min{200, 200} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28		39	30		300	2
Plant B	24		32	26		400	2
Plant C				200		0	
Plant D			200		300	0	
Plant E	0	300				0	
Demand	500	0	0	200	0		
Penalty	4		7	4			

Max{2, 2, 4, 7, 4} = 7.

Min{2, 2} = 2.

Min{300, 0} = 0.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	28		0	30		300	2
Plant B	24			26		400	2
Plant C				200		0	
Plant D			200		300	0	
Plant E	0	300				0	
Demand	500	0	0	200	0		
Penalty	4			4			

Max{2, 2, 4, 4} = 4.

Min{2, 2} = 2.

Min{500, 300} = 300.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	300		0			0	
Plant B	24			26		400	2
Plant C				200		0	
Plant D			200		300	0	
Plant E	0	300				0	
Demand	200	0	0	200	0		

Penalty	0		0			

Max{0, 0, 2} = 2.

Min{0, 0} = 0.

Min{200, 400} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	300		0			0	
Plant B	200			26		200	0
Plant C				200		0	
Plant D			200		300	0	
Plant E	0	300				0	
Demand	0	0	0	200	0		
Penalty				0			

Min{200, 200} = 200.

	Product 1	Product 2	Product 4	Product 4	Pseudo	Supply	Penalty
Plant A	300		0			0	
Plant B	200			200		0	
Plant C				200		0	
Plant D			200		300	0	
Plant E	0	300				0	
Demand	0	0	0	0	0		
Penalty							

e) Using Transportation Simplex method to obtain the next iteration of part (d)

Note that (letting $u_1 = 0$)

$u_1 + v_1 = 28$	$u_1 = 0, v_1 = 28$	$C_{12}^* = U_1 + V_2 - C_{12} = 0 + 32 - 29 = 3$
$u_2 + v_1 = 24$	$u_2 = -4, v_1 = 28$	$C_{14}^* = U_1 + V_4 - C_{14} = 0 + 30 - 30 = 0$
$u_5 + v_1 = 27$	$u_5 = -1, v_1 = 28$	$C_{22}^* = U_2 + V_2 - C_{22} = -4 + 32 - 33 = -5$
$u_5 + v_2 = 31$	$u_5 = -1, v_2 = 32$	$C_{23}^* = U_2 + V_3 - C_{23} = -4 + 39 - 32 = 3$
$u_1 + v_3 = 39$	$u_1 = 0, v_3 = 39$	$C_{31}^* = U_3 + V_1 - C_{31} = -5 + 28 - 30 = -7$
$u_4 + v_3 = 28$	$u_4 = -11, v_3 = 39$	$C_{32}^* = U_3 + V_2 - C_{32} = -5 + 32 - 28 = -9$
$u_2 + v_4 = 26$	$u_2 = -4, v_4 = 30$	$C_{42}^* = U_4 + V_2 - C_{42} = -11 + 32 - 32 = -11$
$u_3 + v_4 = 25$	$u_3 = -5, v_4 = 30$	$C_{44}^* = U_4 + V_4 - C_{44} = 30 - 11 - 32 = -13$
		$C_{53}^* = U_5 + V_3 - C_{53} = -1 + 39 - 31 = 7$
		$C_{54}^* = U_5 + V_4 - C_{54} = -1 + 30 - 31 = -2$

V_j / U_i	28	32	39	30
0	28 | 300	29	39 | 0	30
-4	24 | 200	33	32	26 | 200
-5	30	28	--	25 | 200
-11	--	32	28 | 200	32
-1	27 | 0	31 | 300	31	31

Letting value 31 be the entering variable, the loop will go from C_{54} -> C_{24} -> C_{21} -> C_{51} -> C_{54}. Thus, C_{51} will be the leaving variable. Min{200, 200, 0} = 0. So, the next iteration will be

V_j / U_i	28	32	39	30
0	28 | 300	29	39 | 0	30
-4	24 | 200	33	32	26 | 200
-5	30	28	--	25 | 200
-11	--	32	28 | 200	32
-1	27	31 | 300	31	31 | 0

Problem #3

In a maximization LP problem, the first iteration table is given as:

Basic	X_1	X_2	X_3	S_1	S_2	S_3	S_4	Solution
Z	-8	-20	-10	0	0	0	0	0
S_1	-2	6	1	1	0	0	0	10
S_2	2	-5	1	0	1	0	0	3
S_3	4	4	-2	0	0	1	0	4
S_4	-3	4	5	0	0	0	1	2

Find the second iteration tableau.

Note that X_2 is the most negative column. So, X_2 will be the pivot column.

Also, $minimum\left\{\frac{10}{6}, \frac{4}{4}, \frac{2}{4}\right\} = \frac{2}{4}$ So, Θ = ½. So, the pivot row will be S_4.

Thus, the pivot will be (X_2, S_4) = 4.

Now, recall that pivot equation = Pivot Row / Pivot, i.e. $S_4/4$ -> X_2

Using the pivot equation, the remaining rows may be iterated using the following equation: Old Row – Pivot Column Value * Pivot Equation, i.e. $S_2 – (-5)*X_2 = S_2$. Using these methods yields the following iteration in tabular form.

Basic	X_1	X_2	X_3	S_1	S_2	S_3	S_4	Solution
Z	-23	0	15	0	0	0	5	10
S_1	10/4	0	-26/4	1	0	0	-3/2	7
S_2	¾	0	29/4	0	1	0	5/4	11/2
S_3	7	0	-7	0	0	1	-1	2
X_2	-3/4	1	5/4	0	0	0	1/4	½

Problem #4

Maximize $Z = 2x_1 + x_2 – 3x_3$

Subject to:

$2x_1 + x_2 = 14$

$x_1 + x_2 + x_3 \geq 6$

$x_1, x_2, x_3 \geq 0$

Solve the problem by using the M-technique.

Maximize $Z = 2x_1 + x_2 – 3x_3 – Ma_1 – Ma_2$

Subject to:

$2x_1 + x_2 + a_1 = 14$

$x_1 + x_2 + x_3 - s_1 + a_2 = 6$

$x_1, x_2, x_3, a_1, a_2, s_1 \geq 0$

Let ERO = Elementary Row Operation

ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
1	-2	-1	3	M	0	M	1	0
2	2	1	0	1	0	0	0	14
3	1	1	1	0	-1	1	0	6

ERO: $-M*R_2 + R_1 \rightarrow R_1$

ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
1	-2M – 2	-1 – M	3	0	0	M	1	-14M
2	2	1	0	1	0	0	0	14

3	1	1	1	0	-1	1	0	6

ERO: $-M*R_3 + R_1 \rightarrow R_1$

ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
1	-3M – 2	-1 – 2M	3 – M	0	M	0	1	-20M
2	2	1	0	1	0	0	0	14
3	1	1	1	0	-1	1	0	6

Basic Variables: a_1, a_2, Z

Non-basic Variables: x_1, x_2, x_3, s_1

So, $a_1 = 14$, $a_2 = 6$, $Z = -20M$ & $\{x_1, x_2, x_3, s_1 = 0\}$.

Since the basic variables are non-negative, then a solution exists.

Now, the simplex method may be applied.

BV	ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
Z	1	-3M – 2	-1 – 2M	3 – M	0	M	0	1	-20M
a_1	2	2	1	0	1	0	0	0	14
a_2	3	1	1	1	0	-1	1	0	6

Note that -3M – 2 is the most negative column entry for Row 1. Also, 14/2 = 7; 6/1 = 6. Thus, 1 is the pivot element.

ERO:

$R_3/1 \rightarrow R_3$

$-2R_3 + R_2 \rightarrow R_2$

$(3M + 2)*R_3 + R_1 \rightarrow R_1$

BV	ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
Z	1	0	1 + M	5 + 2M	0	-2M – 2	3M + 2	1	-2M + 12
a_1	2	0	-1	-2	1	2	-2	0	2
x_1	3	1	1	1	0	-1	1	0	6

Note that -2M – 2 is the most negative column entry for Row 1. Also, 2 is the pivot element.

ERO:

$R_2/2 \rightarrow R_2$

$R_2 + R_3 \rightarrow R_3$

$(2M + 2)*R_2 + R_1 \rightarrow R_1$

BV	ROW	x_1	x_2	x_3	a_1	s_1	a_2	Z	Value
Z	1	0	0	3	M + 1	0	M	1	12
s_1	2	0	-1/2	-1	1/2	1	-1	0	1
x_1	3	1	½	0	-1/2	0	0	0	7

So, the optimal solution is Z = 12, s_1 = 1, & x_1 = 7.

e-signature

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